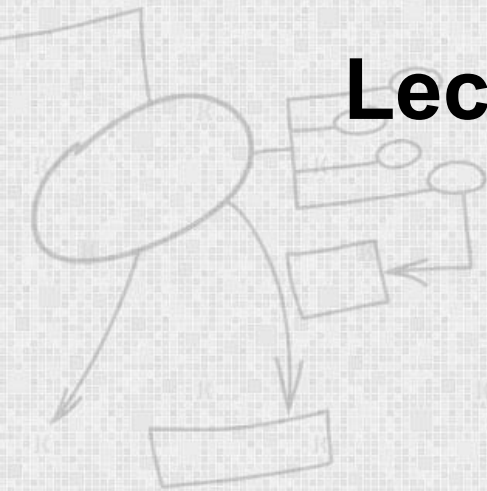


Database Engineering

Lecture 2: Modeling Data in the Organization

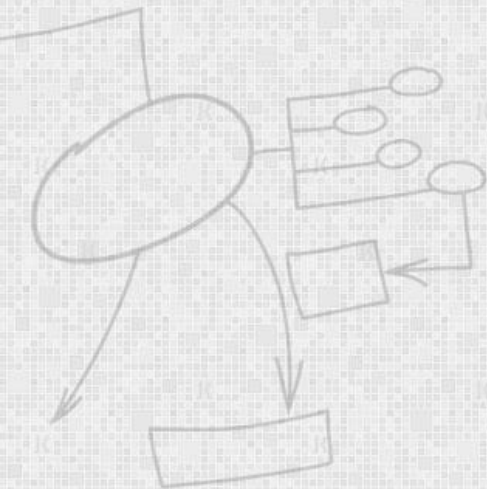
By

Dr Wasi Haider Butt



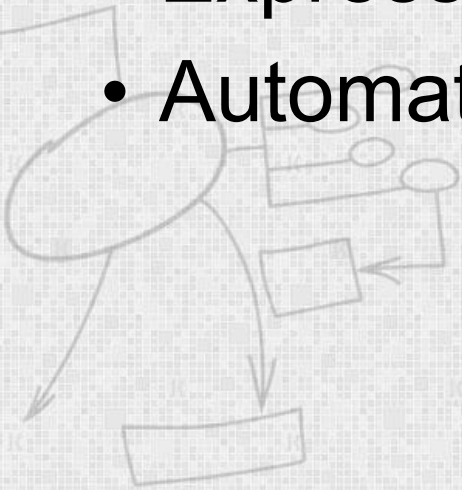
Today's Lecture: Agenda

- **Business Rules**
- **ER Modeling**



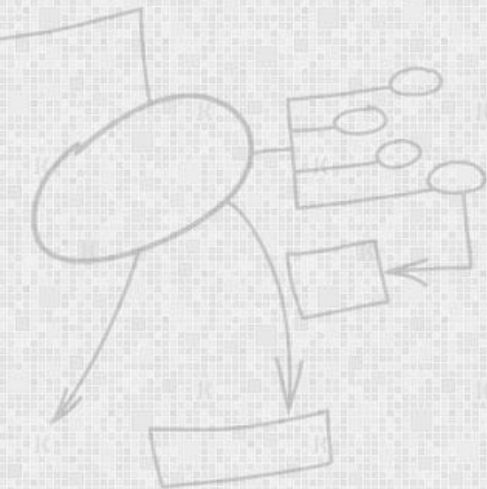
Business Rules

- Statements that define or constrain some aspect of the business
- Assert business structure
- Control/influence business behavior
- Expressed in terms familiar to end users
- Automated through DBMS software



Scope of Business Rules

- We see business rules in Information system context. What all fall inside and what all is outside the scope.



E-R Model Constructs

- **Entities:**
 - Entity instance—person, place, object, event, concept (often corresponds to a row in a table)
 - Entity Type—collection of entities (often corresponds to a table)
- **Relationships:**
 - Relationship instance—link between entities (corresponds to primary key-foreign key equivalencies in related tables)
 - Relationship type—category of relationship...link between entity types
- **Attribute**—property or characteristic of an entity or relationship type (often corresponds to a field in a table)



Basic E-R notation (Figure 3-2)

Entity symbols

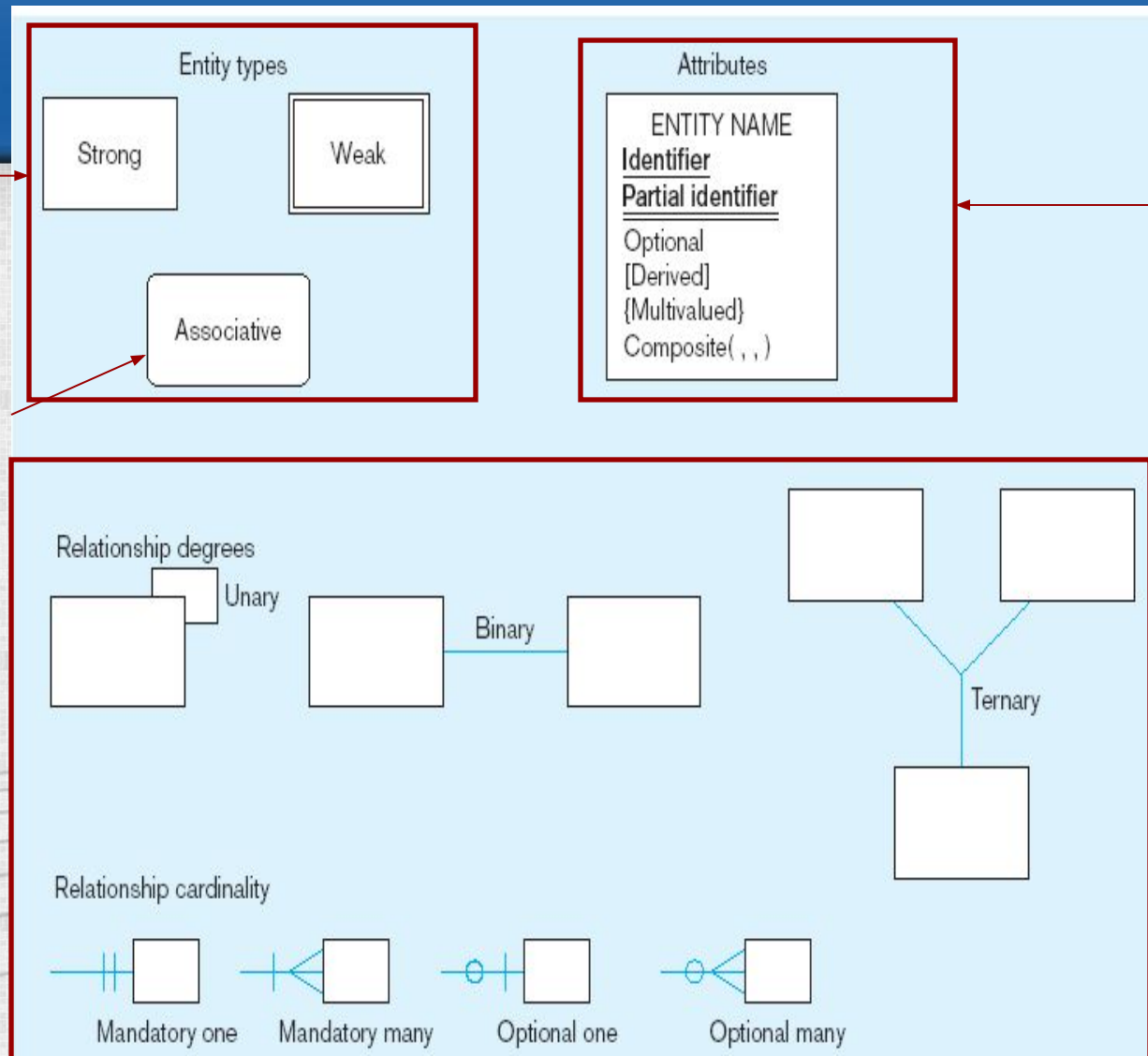
Attribute symbols

A special entity that is also a relationship

Relationship symbols

Relationship degrees specify number of entity types involved

Relationship cardinalities specify how many of each entity type is allowed



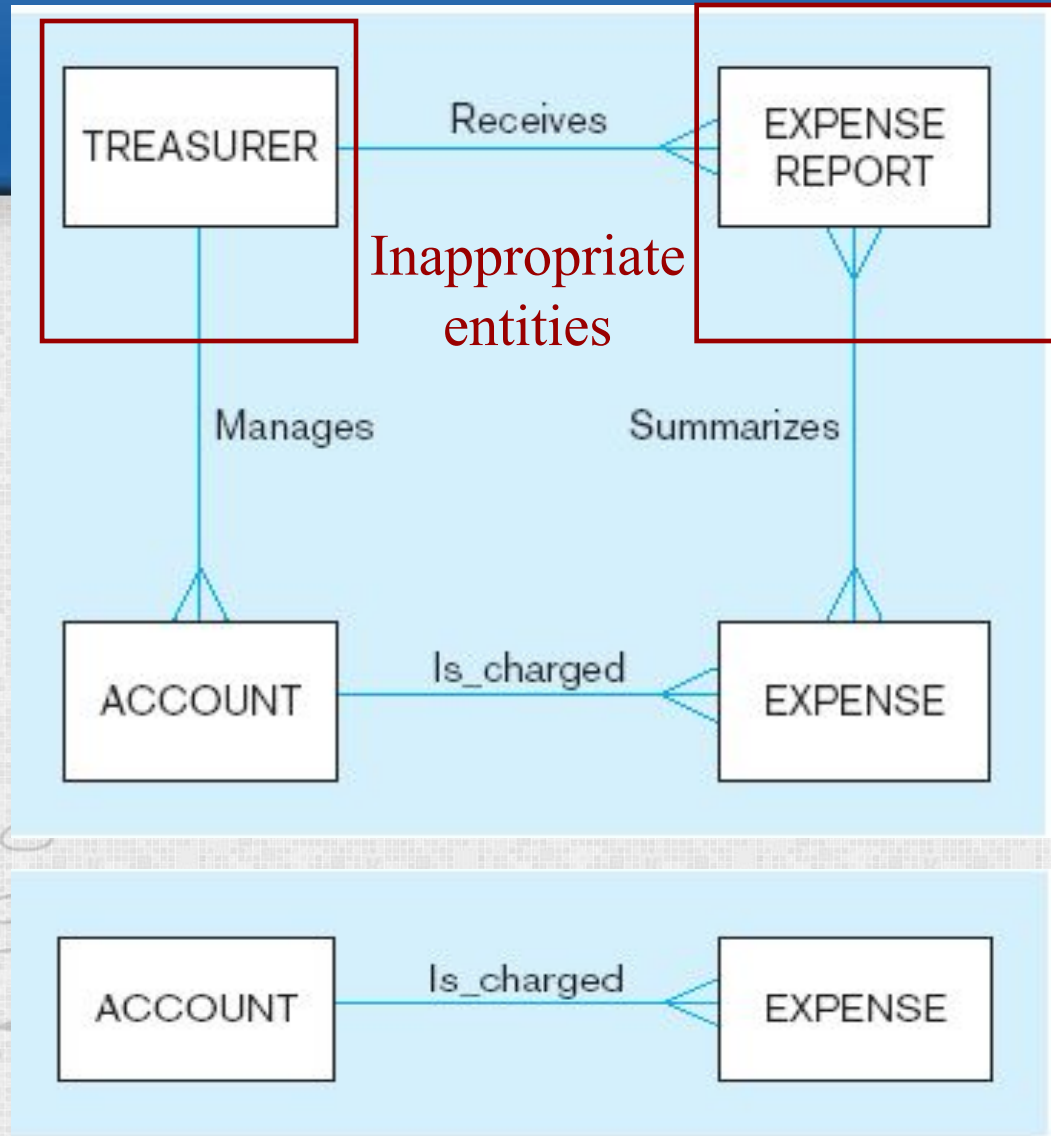
What Should an Entity Be?

- SHOULD BE:
 - An object that will have many instances in the database
 - An object that will be composed of multiple attributes
 - An object that we are trying to model
- SHOULD NOT BE:
 - A user of the database system
 - An output of the database system (e.g., a report)



Figure 3-4 Example of inappropriate entities

**System
user**



**System
output**

**Appropriate
entities**



Attributes

- Attribute—property or characteristic of an entity or relationship type
- Classifications of attributes:
 - Required versus Optional Attributes
 - Simple versus Composite Attribute
 - Single-Valued versus Multivalued Attribute
 - Stored versus Derived Attributes
 - Identifier Attributes



Identifiers (Keys)

- Identifier (Key)—An attribute (or combination of attributes) that uniquely identifies individual instances of an entity type
- Simple versus Composite Identifier
- Candidate Identifier—an attribute that could be a key...satisfies the requirements for being an identifier



Characteristics of Identifiers

- Will not change in value
- Will not be null
- No intelligent identifiers (e.g., containing locations or people that might change)
- Substitute new, simple keys for long, composite keys



Figure 3-7 A **composite** attribute

**An attribute
broken into
component parts**

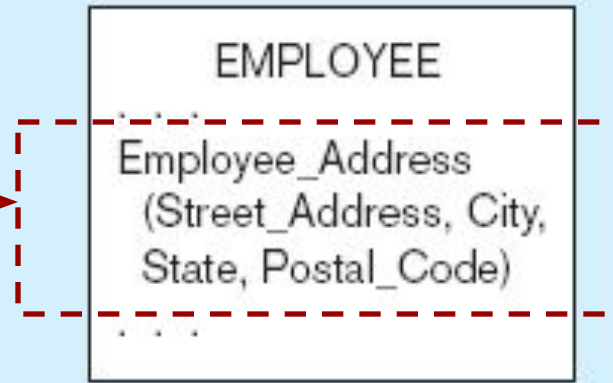
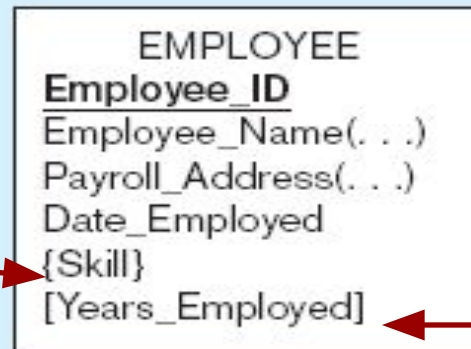


Figure 3-8 Entity with **multivalued** attribute (Skill) and **derived** attribute (Years_Employed)

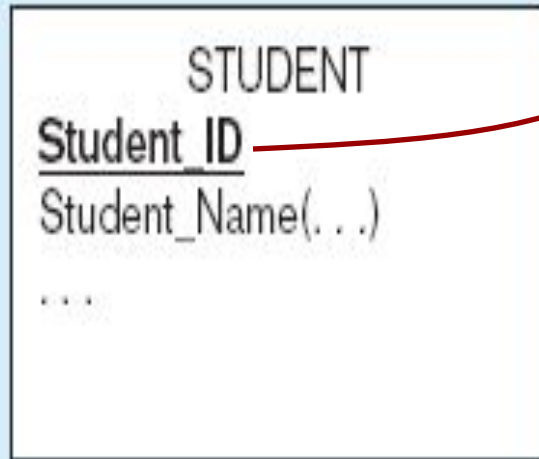
Multivalued
an employee can have
more than one skill



Derived
from date
employed and
current date

Figure 3-9 Simple and composite identifier attributes

The identifier is boldfaced and underlined



(a) Simple identifier attribute



(b) Composite identifier attribute



Figure 3-19 Simple example of time-stamping



This attribute
that is both
multivalued *and*
composite



More on Relationships

- Relationship Types vs. Relationship Instances
 - The relationship type is modeled as lines between entity types...the instance is between specific entity instances
- Relationships can have attributes
 - These describe features pertaining to the association between the entities in the relationship
- Two entities can have more than one type of relationship between them (multiple relationships)
- Associative Entity—combination of relationship and entity

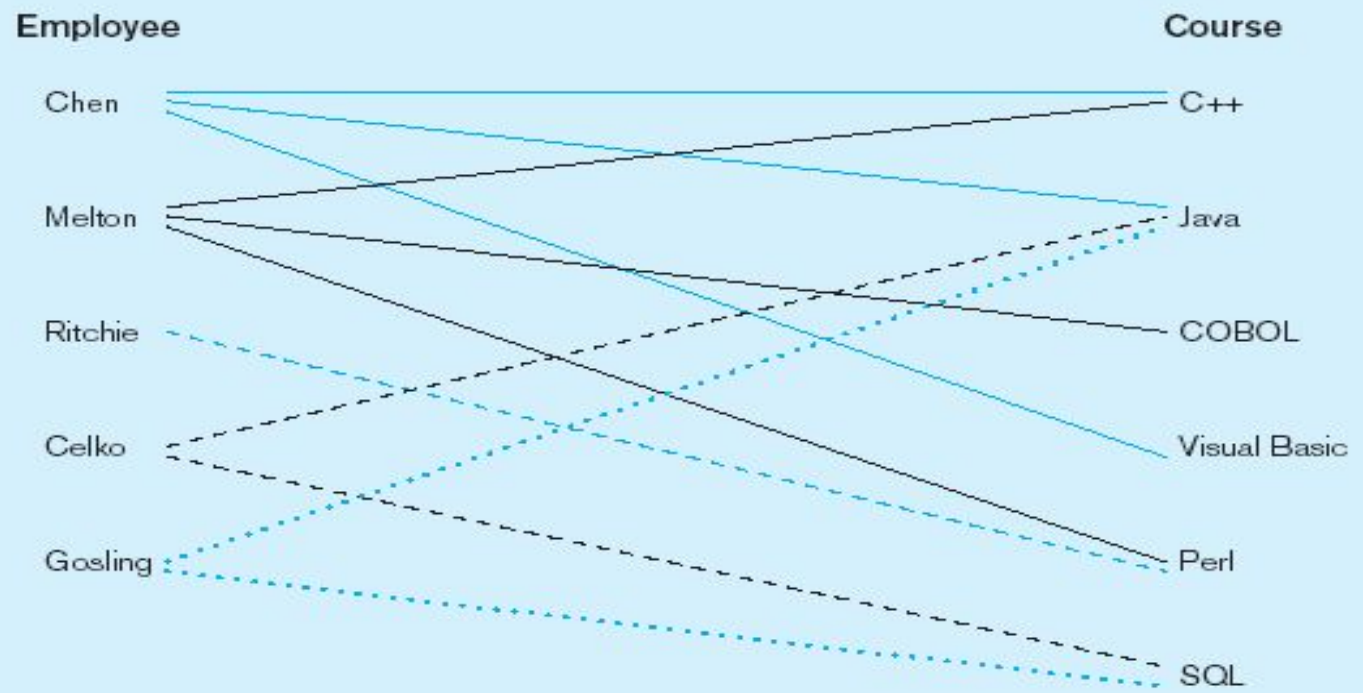


Figure 3-10 Relationship types and instances

a) Relationship type



b) Relationship instances

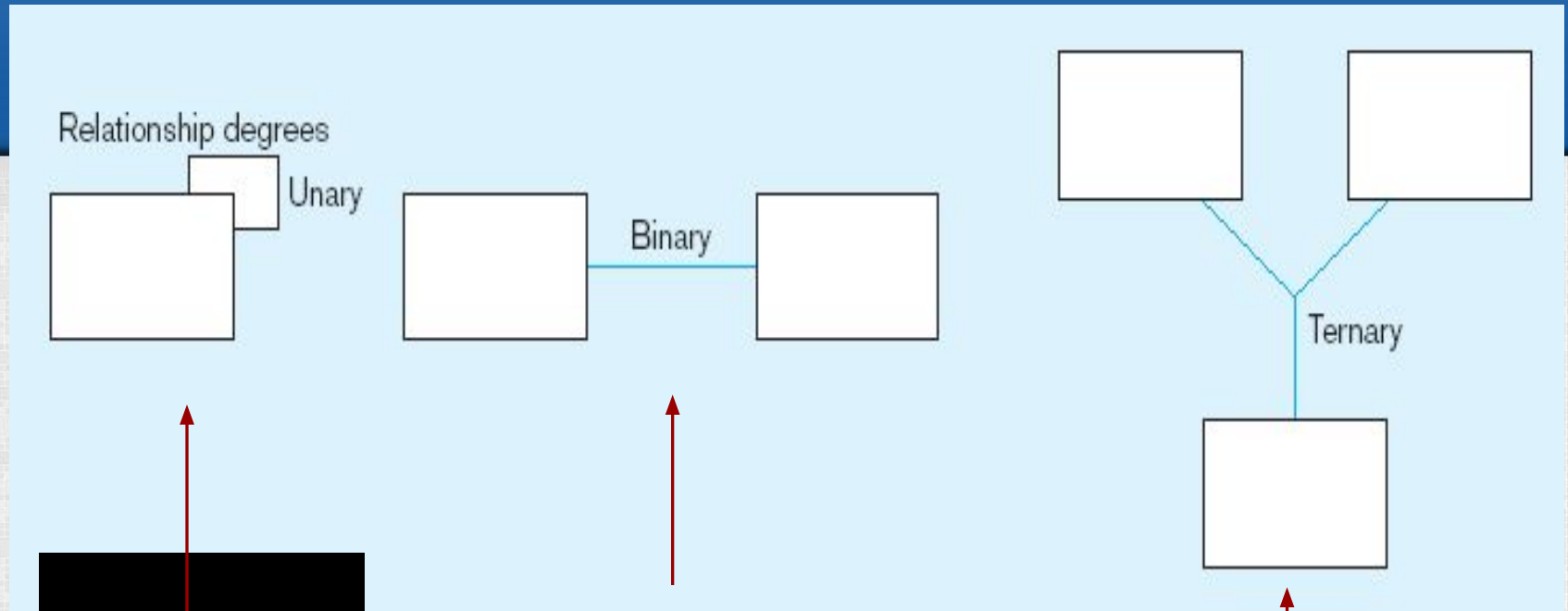


Degree of Relationships

- Degree of a relationship is the number of entity types that participate in it
 - Unary Relationship
 - Binary Relationship
 - Ternary Relationship



Degree of relationships – from Figure 3-2



One entity related to another of the same entity type

Entities of two different types related to each other

Entities of three different types related to each other

Cardinality of Relationships

- One-to-One
 - Each entity in the relationship will have exactly one related entity
- One-to-Many
 - An entity on one side of the relationship can have many related entities, but an entity on the other side will have a maximum of one related entity
- Many-to-Many
 - Entities on both sides of the relationship can have many related entities on the other side



Cardinality Constraints

- Cardinality Constraints - the number of instances of one entity that can or must be associated with each instance of another entity
- Minimum Cardinality
 - If zero, then optional
 - If one or more, then mandatory
- Maximum Cardinality
 - The maximum number



Figure 3-12 Examples of relationships of different degrees

a) Unary relationships

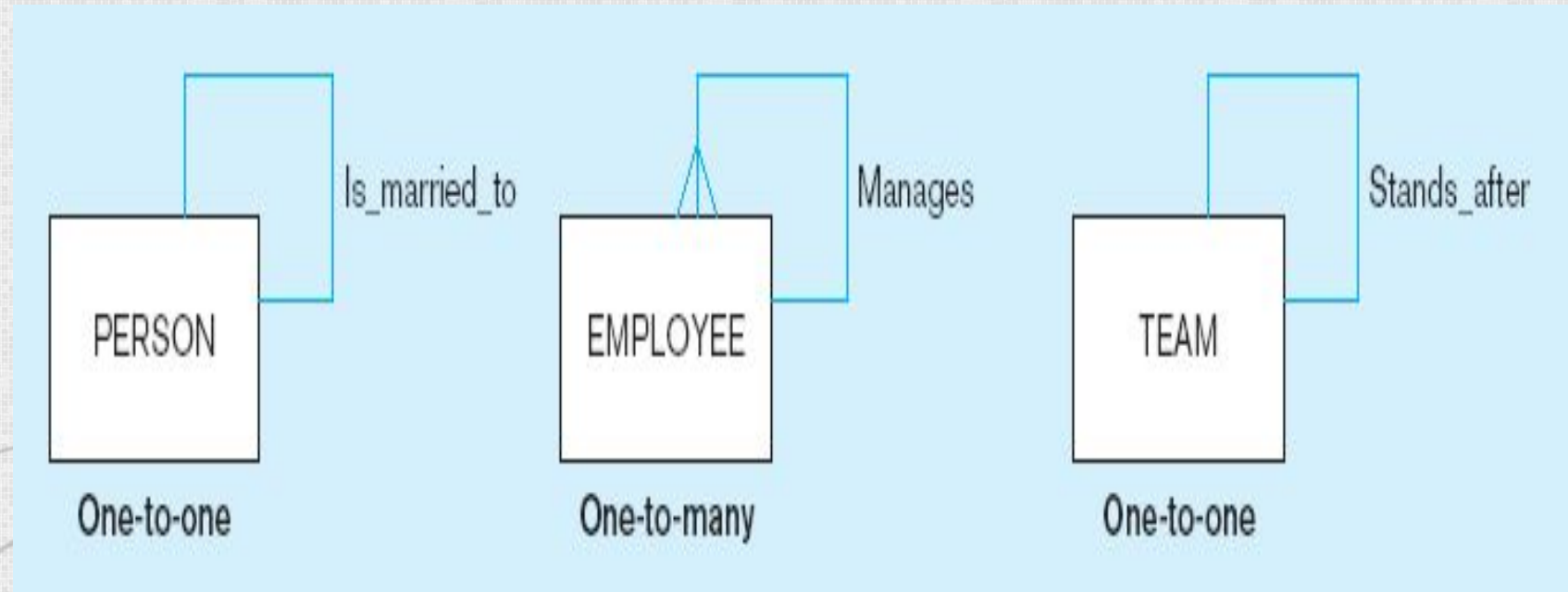


Figure 3-12 Examples of relationships of different degrees (cont.)

b) Binary relationships

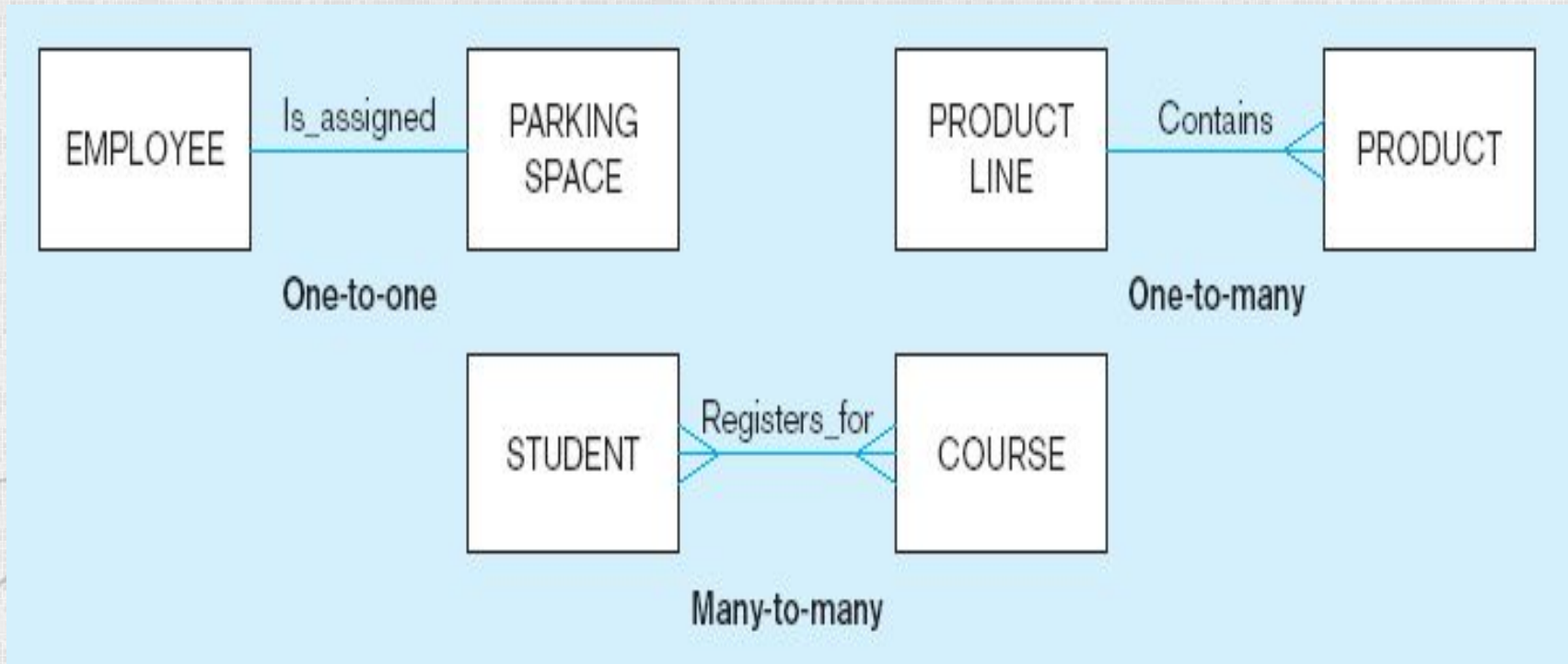
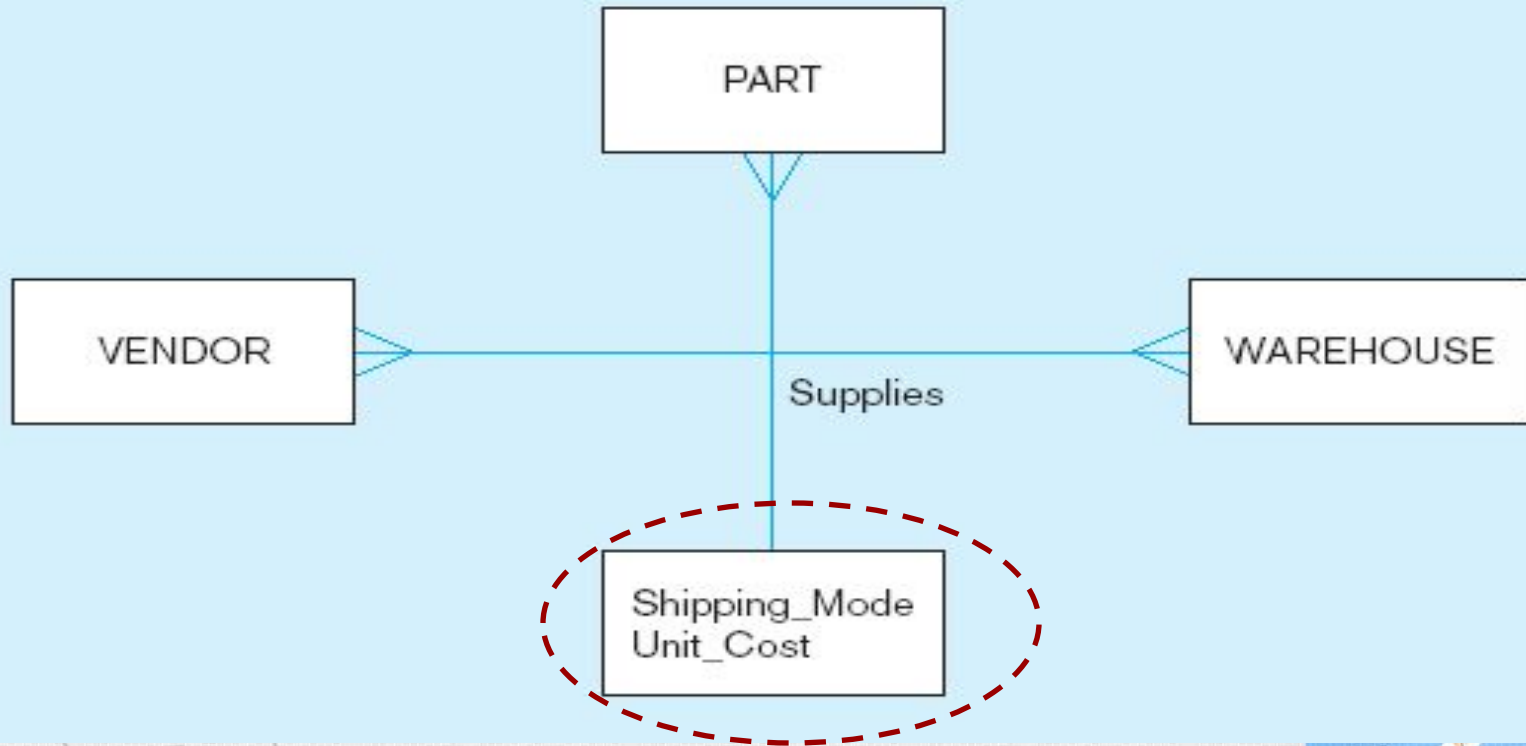


Figure 3-12 Examples of relationships of different degrees (cont.)

c) Ternary relationship

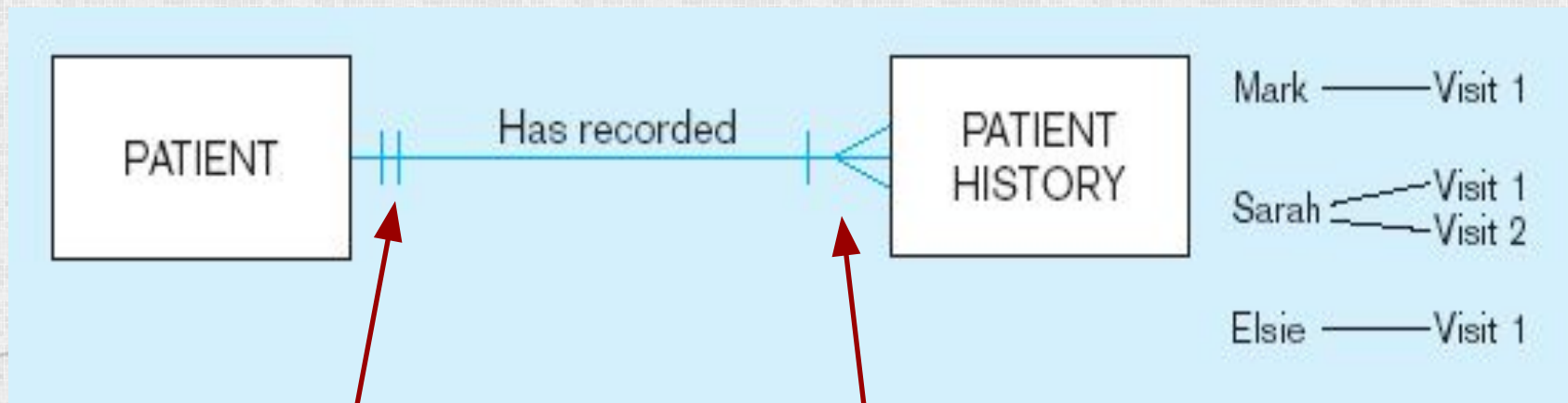
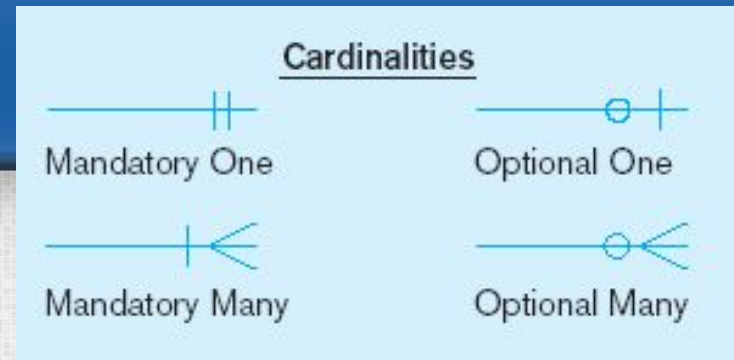


Note: a relationship can have attributes of its own



Figure 3-17 Examples of cardinality constraints

a) Mandatory cardinalities



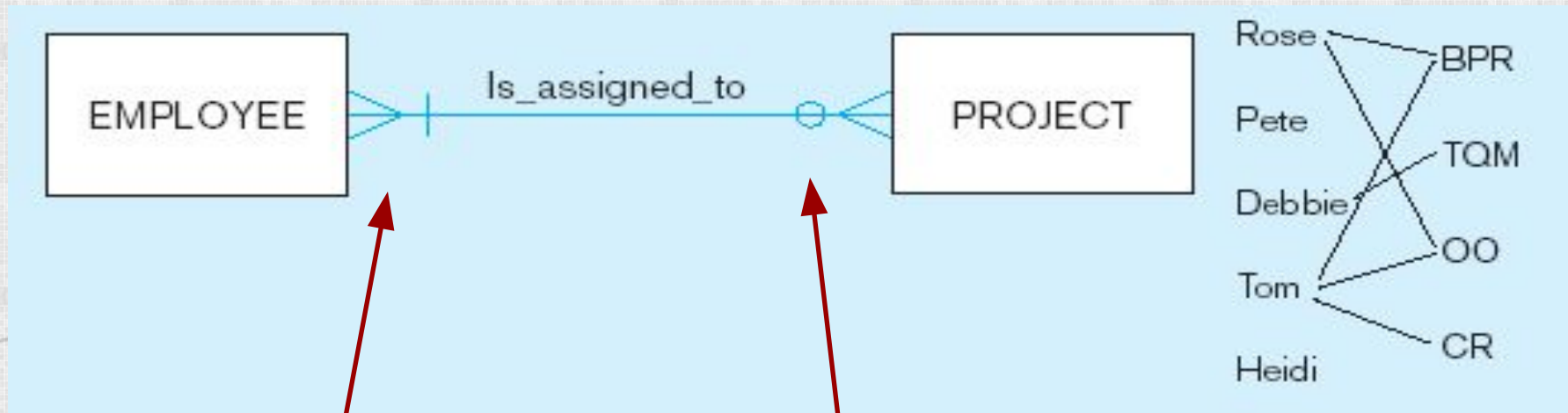
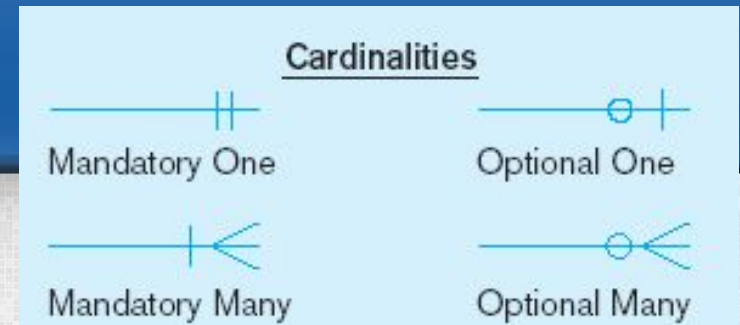
A patient history is recorded for one and only one patient

A patient must have recorded at least one history, and can have many



Figure 3-17 Examples of cardinality constraints (cont.)

b) One optional, one mandatory



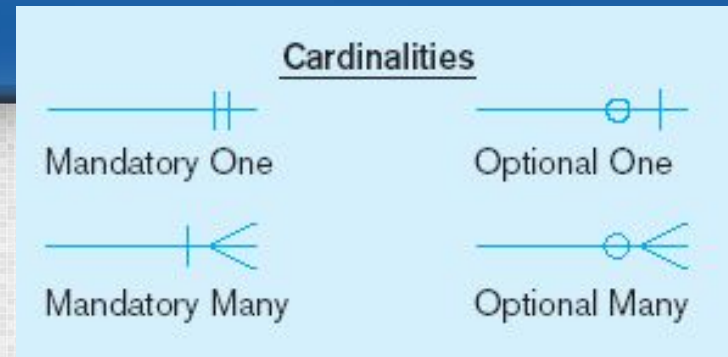
A project must be assigned to at least one employee, and may be assigned to many

An employee can be assigned to any number of projects, or may not be assigned to any at all



Figure 3-17 Examples of cardinality constraints (cont.)

a) Optional cardinalities



A person is married to at most one other person, or may not be married at all

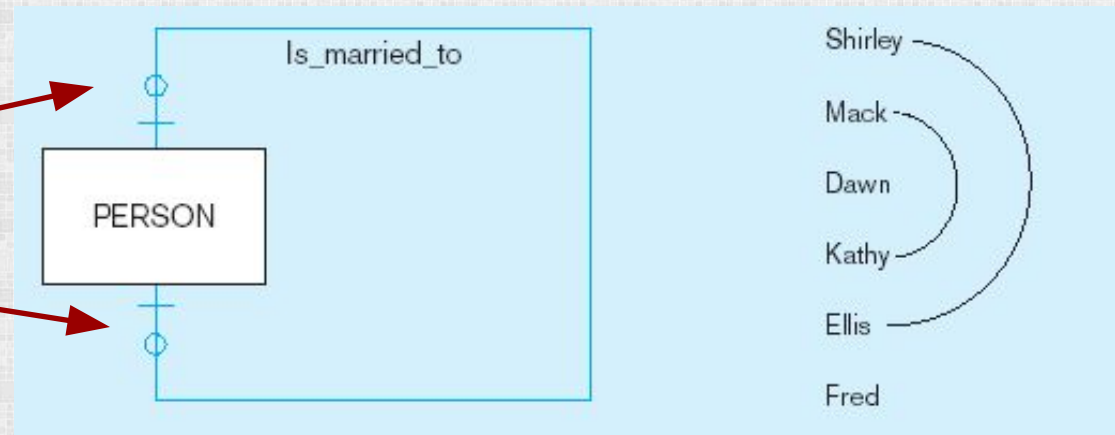
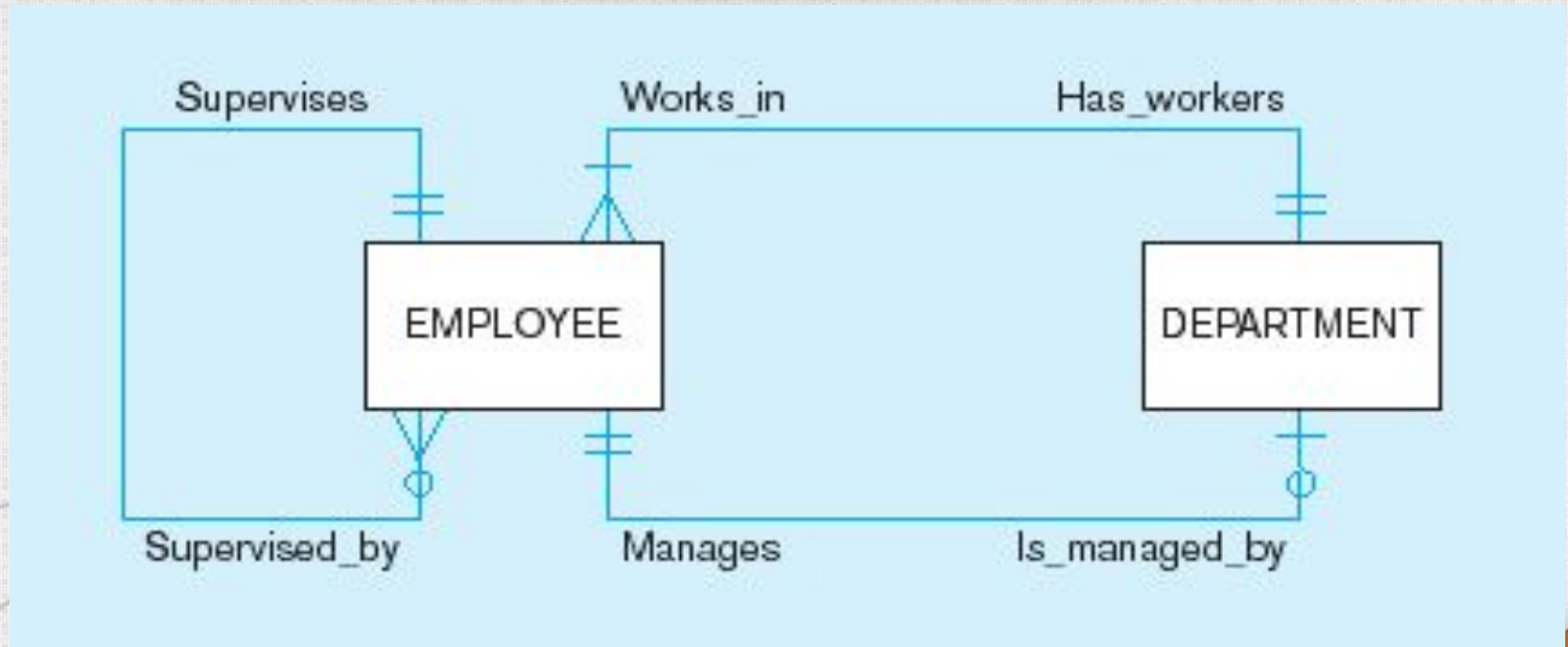


Figure 3-21 Examples of multiple relationships

a) Employees and departments



Entities can be related to one another in more than one way



Strong vs. Weak Entities, and Identifying Relationships

- Strong entities
 - exist independently of other types of entities
 - has its own unique identifier
 - identifier underlined with single-line
- Weak entity
 - dependent on a strong entity (identifying owner)...cannot exist on its own
 - does not have a unique identifier (only a partial identifier)
 - Partial identifier underlined with double-line
 - Entity box has double line
- Identifying relationship
 - links strong entities to weak entities



SUMMARY OF ER-DIAGRAM NOTATION FOR ER SCHEMAS

<u>Symbol</u>	<u>Meaning</u>
	ENTITY TYPE
	WEAK ENTITY TYPE
	RELATIONSHIP TYPE
	IDENTIFYING RELATIONSHIP TYPE
	ATTRIBUTE
	KEY ATTRIBUTE
	MULTIVALUED ATTRIBUTE
	COMPOSITE ATTRIBUTE
	DERIVED ATTRIBUTE



Identifying relationship



Strong entity

Weak entity



Associative Entities

- An **entity**—has attributes
- A **relationship**—links entities together
- When should a *relationship with attributes* instead be an *associative entity*?
 - All relationships for the associative entity should be many
 - The associative entity could have meaning independent of the other entities
 - The associative entity may participate in other relationships other than the entities of the associated relationship
 - Ternary relationships should be converted to associative entities

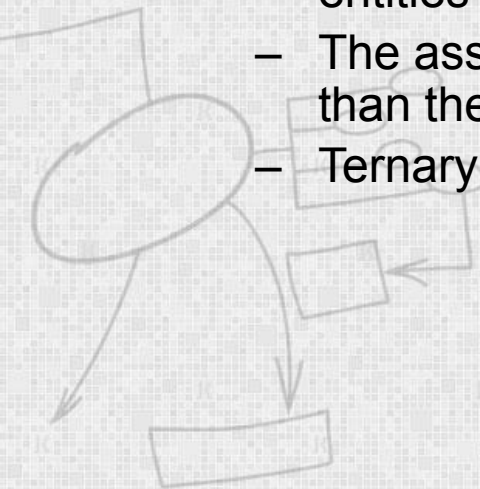
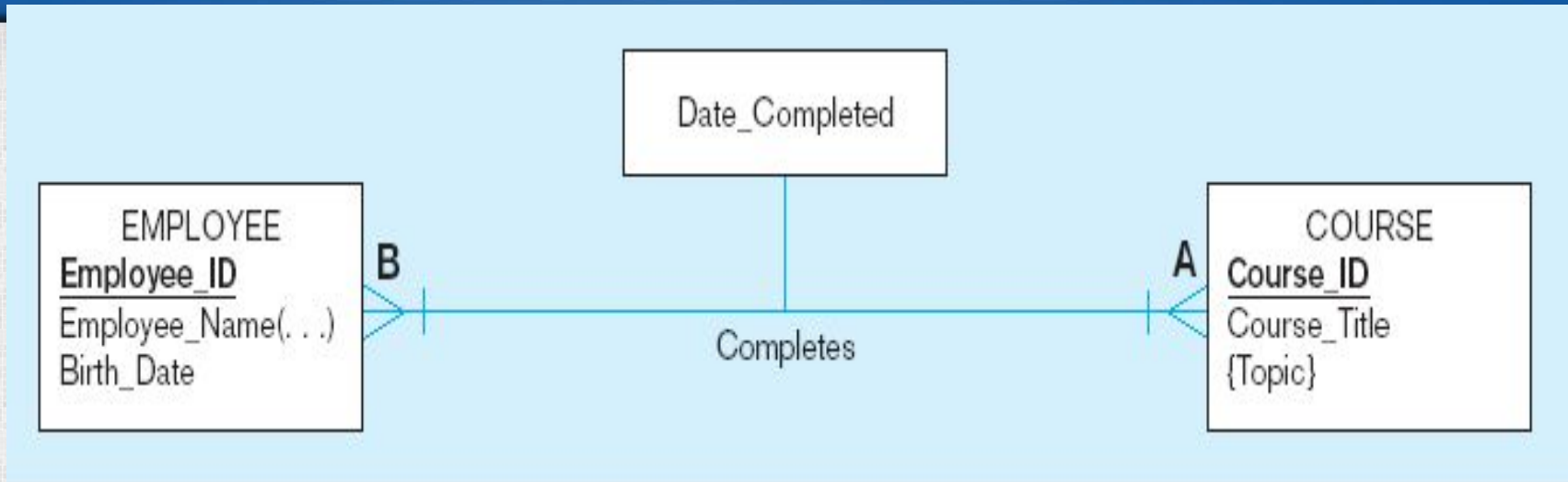


Figure 3-11a A binary relationship with an attribute



Here, the date completed attribute pertains specifically to the employee's completion of a course...it is an attribute of the *relationship*

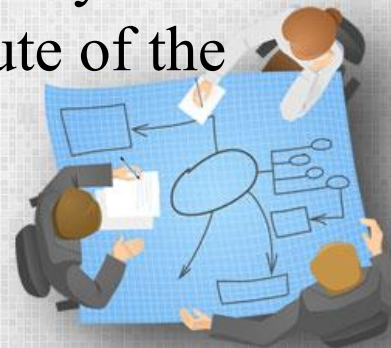
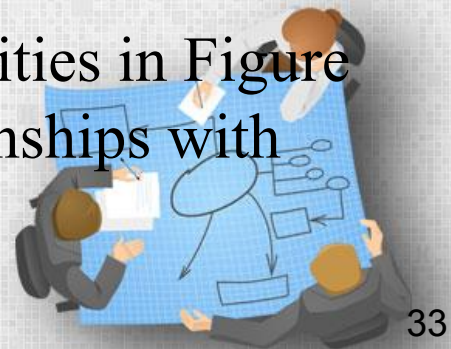


Figure 3-11b An associative entity (CERTIFICATE)



Associative entity is like a relationship with an attribute, but it is also considered to be an entity in its own right.

Note that the many-to-many cardinality between entities in Figure 3-11a has been replaced by two one-to-many relationships with the associative entity.



Creating an ERD from the Investigated Facts

- Identify all the entities.
- Identify all the relationships.
- Identify cardinality and multiplicities (min max).

